

Ankit Shukla

Pune, India | 7972628423 | asankitshukla769@gmail.com | linkedin.com/in/ankit-shukla-47b8a434

SUMMARY

Systems software engineer with 15 years of experience in C/C++ development across Linux, embedded, RTOS, and Windows platforms. Expertise in Linux systems programming, multithreading, concurrency, IPC, memory management, debugging, and performance optimization. Hands-on experience in GPU/graphics driver development across Mesa, LLVM, libdrm, and Linux DRM/KMS, along with multimedia pipeline development (GStreamer, FFmpeg) and embedded, hardware-adjacent software. Experienced technical lead with end-to-end ownership from design through production support.

EDUCATION

Uttar Pradesh Technical University (UPTU) — B.Tech, 2010

Pune University — Diploma in System Software, 2011

EXPERIENCE

Software Engineer 4 — Resource Manager

Dec 2024 – Present

COHESITY INC., Pune

- Develop core C++, Go, and Python software for managing Linux/Windows VMs, disks, and containerized workloads.
- Own design, implementation, automated testing, debugging, and production issue investigation.

Principal Software Engineer

Jan 2023 – Dec 2024

VERITAS TECHNOLOGIES, Pune

- Developed C++ system software for shared licensing and entitlement validation.
- Maintained and validated software across Windows, Linux, AIX, and Solaris.
- Managed platform builds, dependencies, vulnerability remediation, and SDK delivery.

Senior Software Engineer / Technical Lead

Mar 2019 – Jan 2023

SIEMENS MOBILITY, Pune

- Led six engineers developing safety-critical C/C++ software for railway Radio Block Center systems.
- Owned requirements, design, implementation, static analysis, testing, code reviews, release, and mentoring.
- Developed Python automation for engineering and unit-test management.

Specialist — Automotive Embedded Systems

Jun 2015 – Mar 2019

TATA ELXSI, Pune

- Developed C/C++ connectivity software for automotive infotainment systems on QNX and Windows.
- Worked on voice commands, touch/UI input, CAN integration, debugging, and root-cause analysis.
- Built GStreamer- and FFmpeg-based media playback pipelines for infotainment audio/video subsystems.

Senior Software Engineer — Graphics Driver Development

Aug 2014 – Jun 2015

INFOSYS, Hyderabad

- Developed across the AMD Radeon Linux graphics stack using OpenGL, Mesa, LLVM, libdrm, and DRM/KMS.
- Debugged GPU command submission and rendering pipelines; developed OpenGL validation applications.

Software Engineer

Aug 2011 – Aug 2014

ROCKWELL COLLINS, Hyderabad

- Developed C/C++ avionics software on LynxOS.
- Worked with interrupts, MMIO, hardware registers, synchronization, buffer queues, and Trace32 debugging.
- Managed system requirements using DOORS; worked with ARINC653, ARINC429, and DO-178B avionics standards.

TECHNICAL SKILLS

Languages: C, C++ (11/14/17, STL, Templates, RAII, Move Semantics), Python, Go

Systems: Linux, POSIX, IPC, TCP/IP, Multithreading, Synchronization, Memory Management

GPU/Graphics: OpenGL, Mesa, LLVM Graphics Pipeline, libdrm, Linux DRM/KMS, Command Submission

Multimedia: GStreamer, FFmpeg, Audio/Video Codecs, Streaming Pipelines

Embedded: Embedded Linux, QNX, LynxOS, WinCE, ARM, x86, Device Drivers

Avionics/Requirements: DOORS (Requirements Management), ARINC653, ARINC429, DO-178B

Tools: GDB, Trace32/Lauterbach, GoogleTest/GoogleMock, CMake, Bazel, Jenkins, Git